

Panda: A Pretrained Forecast Model for Chaotic Dynamics

Benchmark Investigation and Architectural Analysis

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Outline

- 1 Context: Panda Architecture
- 2 Evaluation Setup
- 3 Where Does Panda Generalize?
- 4 Component Ablations
- 5 Sensor Heterogeneity Bottleneck
- 6 What Is and Is Not Established
- 7 Proposed Directions

What is Panda?

A global pretrained forecast model for chaotic dynamical systems.

Training distribution: 2×10^4 synthetic chaotic ODEs (3–10 dimensional), generated via evolutionary search from 129 founding systems.

Key claim: learns something *universal* about chaotic dynamics; transfers zero-shot to unseen systems and (emergently) to PDEs.

Scale: 21M parameters. Matched against **Chronos 20M** throughout this investigation.

Paper: Lai, Bao, Gilpin. ICLR 2026.

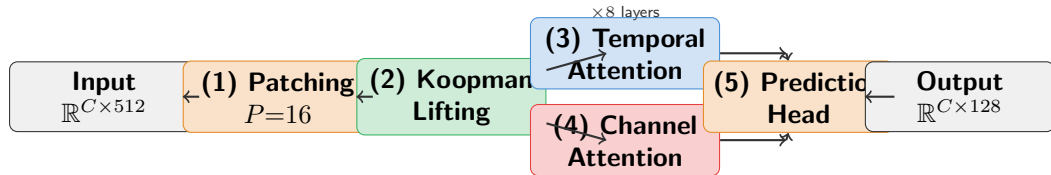
Research Question

Does Panda generalize beyond its chaotic ODE training distribution?

If not: **why not?**

If partially: **which architectural components are responsible?**

Panda Architecture: Five Components



- **(1) Patching:** non-overlapping $P=16$ windows. Theoretically motivated by Takens' embedding theorem.
- **(2) Koopman Lifting:** fixed random polynomial + Fourier features. Approximates eDMD dictionary.

- **(3) Temporal attention:** self-attention over 32 patches per channel. Uses p-RoPE positional encoding.
- **(4) Channel attention:** self-attention over C channels per patch. No positional encoding – permutation equivariant.
- **(5) Prediction head:** linear map to 128-step forecast. Fixed after pretraining.

Why Each Component is Theory-Motivated

Patching $\leftrightarrow$ Takens' Theorem

A patch of P consecutive observations *is* a delay vector. By Takens (1981), for $P \geq 2d + 1$, this is sufficient to reconstruct attractor geometry from a single observable.

Koopman Lifting $\leftrightarrow$ eDMD

Random polynomial and Fourier features form an eDMD dictionary approximating the Koopman operator. Weights are **fixed** at initialization – analogous to reservoir computing. Training them risks overfitting to specific attractors.

Channel Attention $\leftrightarrow$ Dynamical Coupling

Chaotic systems have coupled ODEs. A univariate model forecasting each channel independently may match marginals but fails joint attractor geometry. Channel attention is **permutation equivariant**: channels are an unordered set of state variables, not labelled sensors.

Non-causal Encoder

Attractor geometry is a *global* property of the trajectory. Causal decoders miss it. Processing all 512 context steps simultaneously is the correct inductive bias.

Evaluation Protocol: What Changed and Why

Pre-presentation protocol (flawed):

- **Global normalisation:** statistics computed over full dataset before windowing. Leaks future information.
- **sMAPE as primary metric:** unreliable when values near zero (denominator vanishes on instance-normalised data).
- **$n = 6-10$ windows:** very low statistical power.
- **No significance tests:** differences not distinguished from noise.
- **Single horizon ($H = 128$):** may not generalise.

Fixed protocol (all subsequent experiments):

- **Per-window normalisation:** each context window normalised independently. No future leakage.
- **MAE as primary metric:** reliable on normalised data.
- **$n = 20$ windows:** adequate statistical power.
- **Wilcoxon signed-rank test:** one-sided, per-window. Floor of $p \approx 0.0001$ at $n = 20$.
- **Multiple horizons:** $H \in \{96, 192, 336, 720\}$ matching literature.

Standard Time Series Benchmarks: Fixed Results

Dataset	H	Panda	Chronos	Adv	p
ETTh1	96	0.727	0.663	-0.064	0.844
ETTh1	192	0.819	0.783	-0.036	0.826
ETTh1	336	0.857	0.901	+0.044	0.649
ETTh1	720	0.992	1.019	+0.027	0.774
ETTh2	96	0.874	0.949	+0.076	0.478
ETTh2	336	0.926	1.110	+0.185	0.013*
ETTh2	720	1.114	1.103	-0.011	0.392
Weather	96	0.638	0.812	+0.174	<0.001*
Weather	192	0.722	0.958	+0.236	0.001*
Weather	336	0.848	1.084	+0.236	<0.001*

$n = 20$ windows. * denotes $p < 0.05$ (Wilcoxon signed-rank).

[OBS] Weather: consistent, significant Panda advantage at all three horizons ($p \leq 0.001$). Advantage 0.17–0.24.

[OBS] ETTh1: no significant difference at any horizon. Direction inconsistent.

[OBS] ETTh2: one significant horizon ($H=336$). Three other horizons not significant and inconsistent in direction.

Key open question: what drives the Weather advantage?

Chaotic Systems: Lorenz Rho Sweep

ρ	Regime	Panda	Chronos	Adv
10	Periodic	0.02	0.02	≈ 0
15	Periodic	0.04	0.04	≈ 0
20	Near trans.	0.09	0.31	+0.22
24	Near trans.	0.10	0.44	+0.34
24.74	Onset chaos	0.18	0.65	+0.47
28	Chaotic	0.25	0.89	+0.64
60	Chaotic	0.48	1.17	+0.69

Caveat: Lyapunov estimator (Rosenstein method) overestimates. Literature value for $\rho = 28$: $\lambda_1 \approx 0.9$; estimator returned 1.66. X-axis not quantitatively reliable.

[PAT] Panda advantage is approximately zero at $\rho = 10, 15$. Steps up near the bifurcation at $\rho = 24.74$. Remains positive and growing through the chaotic range.

Alternative explanations not ruled out:

- 1 Signal statistics (variance, spectrum) change discontinuously at bifurcation. Any model suited to high-variance aperiodic signals would show this.
- 2 Chronos tokenisation fails on aperiodic continuous signals independently of Panda.

Complexity Continuum: Non-Monotone Result

System	Regime	Panda	Chronos	Adv (p)
Harmonic	Periodic	0.065	0.435	+0.370 (0.004*)
Van der Pol	Limit cycle	0.032	0.043	+0.011 (0.027*)
Duffing	Weakly chaos	0.581	0.795	+0.214 (0.055)
Rosler	Chaotic	0.083	0.387	+0.304 (0.004*)
Lorenz	Chaotic	0.056	0.531	+0.476 (0.004*)

$H = 96$, $n = 8$ windows.

Critical anomaly:

[OBS] Harmonic advantage (+0.370) is second largest. Larger than weakly chaotic Duffing (+0.214).

[OBS] Van der Pol (limit cycle): smallest advantage (+0.011), anomalously low.

Directly contradicts the chaos-specific hypothesis.

PDE Generalization: Burgers Viscosity Sweep

ν	Regime	Panda	Chronos	Adv	p
2.000	Diffusion	0.015	0.019	+0.004	0.273
1.000	Diffusion	0.019	0.057	+0.038	0.004*
0.500	Transition	0.040	0.102	+0.062	0.004*
0.100	Wk. nonl.	0.113	0.224	+0.111	0.004*
0.050	Shock form.	0.137	0.287	+0.149	0.004*
0.020	Str. shock	0.139	0.259	+0.120	0.004*
0.010	Str. shock	0.138	0.233	+0.095	0.004*
0.005	Str. shock	0.151	0.273	+0.122	0.004*

16 PCA channels, $n = 8$ windows, $H = 128$.

Key finding:

[OBS] Panda wins significantly at $\nu = 1.0$ (diffusion-dominated, physically non-chaotic, $\lambda_1 \leq 0$). This is inconsistent with a chaos-specific mechanism.

[PAT] Advantage increases monotonically from $\nu = 1.0$ to $\nu = 0.05$, after which the advantage in non-monotone (though positive)

Competing explanations:

- 1 Channel attention captures PDE spatial coupling present even in non-chaotic flows.
- 2 Chronos tokenisation fails on PCA modal time series regardless of chaoticity.

Experiment Map: Which Component Does Each Experiment Target?

Component	Experiments	Result
Channel attention	Exp 9 (Weather univariate ablation) Exp 27 (Burgers univariate ablation) Exp 22 (node embedding proxy)	Not the driver of Weather advantage Not the driver of Burgers advantage Marginal directional effect only
Prediction head	Exp 23 (head fine-tuning)	Fine-tuning at H=96 hurts (-0.091 MAE)
Koopman lifting	Exp 21 (permutation entropy) Topology analysis Exp 19 (Harmonic oscillator)	PE at order 3 fails as discriminator in all contexts Weather $\approx$ Lorenz $\neq$ ETTh (estimator unreliable) Large advantage on pure sinusoid: unexplained
Temporal attention	Not ablated	Requires code-level access. Gap in investigation.
Patching	Exp 19 (complexity continuum)	Partial: periodic regime, non-monotone pattern
Pretraining dist.	Exp 3, 19	ETTh gap consistent with distribution mismatch

Reading this table: every row is a component; every finding is a correlation, not a causal result. The positive mechanism behind any advantage has not been isolated.

Channel Attention: Weather Univariate Ablation (Exp 9)

Question: does channel attention drive the Weather advantage?

Method: compare multivariate Panda (all 21 channels jointly) vs univariate Panda (each channel independently – suppresses cross-channel attention). $n = 8$, $H \in \{96, 336\}$.

H	Uni MAE	Multi MAE	Δ	p
96	0.554	0.611	-0.057	0.074
336	0.847	0.876	-0.030	0.371

$\Delta = \text{Uni} - \text{Multi}$. Negative means univariate has lower MAE.

[OBS] Univariate Panda has lower MAE than multivariate at both horizons. Neither result is statistically significant at $n = 8$.

What this establishes: channel attention does not improve Weather performance and may marginally hurt it.

What this does not establish: whether the temporal architecture alone is the driver (no temporal-only ablation conducted).

Confidence: Medium for the observation. The effect is in a consistent direction.

Channel Attention: Burgers Univariate Ablation (Exp 27)

Question: does channel attention drive the non-chaotic Burgers advantage?

Method: multivariate vs univariate Panda at $\nu \in \{2.0, 1.0, 0.5\}$ (non-chaotic and transition regimes). $n = 8$, $H = 96$.

ν	Multi	Uni	Adv multi	Adv uni
2.0	0.0094	0.0078	+0.006 (ns)	+0.001 (ns)
1.0	0.0165	0.0126	-0.002 (ns)	+0.003*
0.5	0.0235	0.0241	+0.033*	+0.023*

ns = not significant.

[OBS] At $\nu = 1.0$: multivariate Panda advantage is not significant; univariate Panda advantage is. Univariate outperforms multivariate.

[OBS] At $\nu = 0.5$: both significant; essentially equivalent performance.

[PAT] Channel attention is not the driver of Panda's advantage on Burgers at non-chaotic viscosities. This mirrors the Weather result.

Cross-experiment pattern: univariate Panda is competitive or better in *both* Weather and non-chaotic Burgers.

Prediction Head: Fine-Tuning (Exp 23)

Question: is the prediction head (pretrained on chaotic ODE distributions) a recoverable bottleneck for Weather?

Method: fine-tune only the prediction head on Weather for 50 gradient steps. Encoder frozen. Compare against base Panda.

H	Base	FT	Δ Panda	Adv FT
96	0.609	0.699	-0.091	+0.071
336	0.870	0.881	-0.012	+0.198

$\Delta < 0$ means fine-tuning *hurts*. Advantage = Chronos MAE - Panda MAE.

[OBS] At $H=96$: head fine-tuning worsens Panda MAE by 0.091 ($\approx 15\%$ relative increase). A substantial degradation.

[OBS] At $H=336$: small degradation (-0.012).

Competing explanations:

- 1 Encoder and head co-adapted during pretraining; fine-tuning the head alone introduces mismatch.
- 2 50 steps sufficient to corrupt, not sufficient to converge to a new optimum.
- 3 Head is not the bottleneck; representation upstream is.

Conclusion: lightweight head-only adaptation is not a viable path.

Koopman Lifting: Permutation Entropy Analysis (Exp 21)

Question: can permutation entropy (PE) at order 3 predict Panda's advantage across systems?

Result: PE at order 3 fails on three independent grounds.

Dataset group	PE range	Adv range
Burgers PCA	0.024–0.132	0.004–0.149
Lorenz sweep	0.460–0.494	0–0.69
ETT/Weather	0.756–0.955	–0.06–0.24

These three groups occupy disjoint PE scales with no overlap.

[OBS] Lorenz PE values are nearly identical from $\rho = 10$ (periodic) to $\rho = 60$ (strongly chaotic): PE varies by < 0.05 while Panda advantage varies by 0.69. The metric cannot discriminate.

[OBS] Burgers PE is an order of magnitude lower than real datasets; operating in an incomparable numerical regime.

[PAT] PE at order 3 is not a useful predictor. Higher order ($m = 5, 6$) on longer time series would partially address within-group discrimination, but cannot fix the between-group scale problem without normalisation.

Topology Analysis

Question: does geometric similarity to Panda's training distribution predict where it generalises?

Method: Grassberger-Procaccia correlation dimension estimated per dataset.

Dataset	Corr. dim	Panda wins?
Weather	0.889	Yes (significant)
Lorenz $\rho=28$	0.862	Yes
Burgers $\nu=0.05$	0.741	Yes
ETTh1	1.541	No
ETTh2	1.617	No

[PAT] Datasets where Panda wins (Weather, Lorenz, Burgers) all have correlation dimension < 1.0 ; datasets where Panda does not win (ETTh1, ETTh2) have dimension > 1.5 .

Critical caveat: the Grassberger-Procaccia estimator is known to be unreliable on multivariate real-world data with limited samples. This is a single noisy estimate per dataset.

Status: suggestive pattern, insufficient reliability to act on. Persistent homology on Vietoris-Rips filtrations would provide a more rigorous test.

Decomposition Story: Three Experiments, Revised Conclusion

Question: does Panda's Weather advantage come from better handling of the periodic/trend component?

Exp 11 (naive FFT projection):

- Decomp collapses Panda's advantage on Weather.
- Panda MAE +11% after decomp.
- *Interpretation at the time:* Panda needs full mixed signal.

Exp 17 (improved period projection):

- Same result. Advantage collapses/reverses.
- Weather H=96: $+0.159 \rightarrow -0.038$.
- Confirms Exp 11 not an artifact of naive projection.

Exp 20 (Chronos residual ablation – critical):

- Chronos *also* degrades after decomposition.
- Chronos MAE $+0.277$ at H=96 (significant, $p = 0.016$).
- Comparable magnitude to Panda degradation.

Revised conclusion:

[OBS] Both models degrade approximately equally when the periodic component is removed. The earlier interpretation (Panda specifically depends on full signal) must be revised. FFT projection error likely degrades both models proportionally. The decomposition result is **not diagnostic** of Panda-specific signal dependence.

Motivation: Why Sensor Heterogeneity?

Panda's pretraining: channels are *state variables* of a single dynamical system. They are homogeneous by construction – all coordinates of the same ODE.

Weather dataset: 21 channels measuring temperature, humidity, pressure, wind speed, dew point ... Each sensor has qualitatively different dynamics.

Hypothesis: Panda's permutation-equivariant channel attention was trained assuming channels are interchangeable. On heterogeneous sensors, importing coupling patterns between physically unrelated channels may be harmful.

Design: three experiments, each adding one additional control.

Heterogeneity Index

Per-channel dynamical feature vectors (std, lag-1 autocorrelation, spectral entropy, dominant frequency) used to cluster Weather channels.

Heterogeneity index $\in [0, 1]$: average pairwise feature distance between channels in a subset.

Three subsets: **homogeneous** (same cluster), mixed (two clusters), **heterogeneous** (maximally distinct clusters).

Heterogeneity Exp 1: Stratification (Exp 24)

Subset	Het	H	Panda	Chronos	Adv (p)
Homogeneous	0.035	96	0.318	0.689	+0.371 (0.004*)
Homogeneous	0.035	336	0.817	1.183	+0.365 (0.004*)
Mixed	0.668	96	0.581	0.786	+0.205 (0.039*)
Mixed	0.668	336	0.925	0.982	+0.056 (0.371)
Heterogeneous	0.947	96	0.618	0.697	+0.079 (0.074)
Heterogeneous	0.947	336	1.123	1.163	+0.040 (0.191)

7 channels per subset, $n = 8$ windows.

[OBS] Panda advantage drops $\approx 5\times$ from homogeneous (+0.371) to heterogeneous (+0.079) at $H=96$.

[OBS] On the heterogeneous subset, Panda advantage is not statistically significant at either horizon.

Confound: homogeneous channels may simply be easier to forecast. Next experiment controls for this.

Heterogeneity Exp 2: Difficulty-Matched Control (Exp 25)

Control: match subsets on per-channel Chronos univariate MAE (individual difficulty).

Subset	Het	H	Panda	Chronos	Adv (p)
Homo matched	0.086	96	0.331	0.700	+0.369 (0.004*)
Homo matched	0.086	336	0.841	1.147	+0.305 (0.004*)
Mixed matched	0.602	96	0.475	0.808	+0.333 (0.004*)
Mixed matched	0.602	336	0.856	0.946	+0.090 (0.020*)
Heterogeneous	0.947	96	0.618	0.836	+0.218 (0.074)
Heterogeneous	0.947	336	1.123	1.243	+0.120 (0.125)

[PAT] After controlling for channel-level difficulty, Panda MAE increases monotonically with heterogeneity (0.331, 0.475, 0.618 at $H=96$). Chronos MAE is approximately constant (0.700, 0.808, 0.836).

[OBS] The heterogeneity effect cannot be explained by channel difficulty. Panda specifically degrades; Chronos does not.

Heterogeneity Exp 3: Variance-CV Control + Chronos Calibration (Exp 26)

Second-order control: also match coefficient of variation of difficulty across subsets.

Subset	Het	H	Panda MAE
Hetero controlled	0.853	96	0.605
Hetero controlled	0.853	336	0.867

Compare: homo matched Panda MAE at $H=96$: **0.331**.

Chronos calibration: Chronos MAE across matched subsets:

- homo: 0.776 / hetero: 0.687 (at $H=96$)
- No monotone trend. Chronos insensitive to heterogeneity.

[OBS] After controlling for mean difficulty AND difficulty variance, Panda MAE nearly doubles ($0.331 \rightarrow 0.605$) from homogeneous to heterogeneous. Chronos does not show this pattern.

[PAT] Convergent evidence from three experiments (Exp 24, 25, 26): sensor heterogeneity is a specific architectural bottleneck for Panda, not shared by the univariate Chronos baseline.

This is the most controlled finding in the investigation.

Node Embedding Proxy (Exp 22)

Motivation: if sensor identity is missing, a scalar per-channel offset should help on heterogeneous but not homogeneous subsets.

Method: add per-channel scalar offset (context window mean per channel) before presenting to Panda.

Subset	H	Base	Offset	Δ
Homo matched	96	0.328	0.412	-0.085
Homo matched	336	0.839	0.882	-0.043
Hetero ctrl	96	0.605	0.568	+0.037
Hetero ctrl	336	0.867	0.870	-0.003

$\Delta < 0$ means offset hurts Panda.

[OBS] Direction is correct at $H=96$: offset hurts on homogeneous (-0.085), marginally helps on heterogeneous ($+0.037$).

[OBS] Effect sizes are small. The largest effect (-0.085 on homo) is in the wrong direction.

Competing explanations:

- 1 Scalar offset is too minimal a proxy for learned node embeddings (a vector $\mathbf{e}_i \in \mathbb{R}^d$ per sensor).
- 2 Channel identity is not the missing ingredient; graph interaction structure is.
- 3 Per-window normalisation already removes per-channel mean; double correction introduces noise.

Summary: What the Evidence Shows

High-confidence observations:

- 1 Panda has significant Weather advantage at H=96/192/336 ($n = 20$, $p \leq 0.001$, adv 0.17–0.24).
- 2 Channel attention is not the driver: univariate Panda competitive or better on both Weather and Burgers.
- 3 Harmonic oscillator advantage (+0.370, $p = 0.004$) directly contradicts chaos-specific hypothesis.
- 4 Head fine-tuning at H=96 hurts (-0.091). Not a recoverable bottleneck cheaply.
- 5 Permutation entropy order 3 fails as a discriminator in all tested contexts.
- 6 Sensor heterogeneity is a specific Panda bottleneck (3 converging experiments, difficulty and CV controlled).

Not established:

- 1 **Positive mechanism behind Weather advantage.** Channel attention ruled out. Head ruled out. Koopman lifting and patching not yet ablatable (require code access).
- 2 **Whether chaos is sufficient** for the advantage. Harmonic and Burgers non-chaotic results partially contradict this.
- 3 **Whether sensor identity or interaction structure** is the heterogeneity bottleneck. Node embedding proxy experiment is inconclusive.
- 4 **Whether any intervention improves generalisation.** All experiments are diagnostic. No proposed fix has been shown to recover performance.

Honest Assessment: What We Can and Cannot Claim

What we can claim with confidence:

- Panda generalizes to Weather and Burgers beyond its chaotic ODE distribution.
- It does not generalize well to ETTh1/ETTh2.
- Sensor heterogeneity is a specific failure mode on Weather, not explained by difficulty.
- Channel attention is not the positive mechanism. The head is not cheaply recoverable.
- Panda architecture has at least one untested positive component (Koopman lifting or temporal attention).

What we cannot yet claim:

- *Why* Panda beats Chronos on Weather. No component has been causally implicated.
- Whether any architectural modification would improve generalization. All evidence is correlational.
- Whether the topology analysis (correlation dimension clustering) reflects a real geometric difference or estimator noise.
- Whether the λ_1 -advantage Spearman correlation is real (sequential system addition in Exp 4 was a p-hacking issue).

Proposed Direction 1: Koopman Lifting Ablation

Scientific question: is the Koopman lifting (random polynomial + Fourier features) the positive mechanism behind Panda's advantage?

Why it matters: channel attention is ruled out. Prediction head is ruled out. Temporal attention and Koopman lifting are the remaining untested candidates.

Proposed experiment:

- 1 Replace the lifting with a *learned* linear projection of the same output dimension (512).
- 2 Evaluate on Weather, ETTh1, Burgers at representative horizons.
- 3 If advantage collapses: lifting is the positive mechanism.

Requirement

Requires code-level access to Panda's architecture (not just inference). This is the first positive causal experiment we have designed.

All previous ablations are *inference-time* ablations. The lifting cannot be removed without modifying the model.

What this would establish: the first causal evidence for any architectural component rather than a correlational observation.

Proposed Direction 2: G-SWaN-Style Node Embeddings

Motivation: three controlled experiments show sensor heterogeneity is a specific Panda bottleneck. Prabowo et al. (G-SWaN) propose per-sensor node embeddings with a spatial graph transformer, motivated by exactly the observation that real-world sensors are not interchangeable.

What we tested (Exp 22): scalar per-channel offset. Too minimal; results inconclusive.

Proposed intervention:

Inject a learned vector $\mathbf{e}_i \in \mathbb{R}^d$ per sensor into Panda's channel attention computation. Evaluate whether the heterogeneous-homogeneous MAE gap decreases.

Falsification Criterion

If the homo–hetero Panda MAE gap is *unchanged* after adding proper node embeddings, the bottleneck is **interaction structure** (graph topology) rather than channel identity.

In that case: full graph-attention modification (G-SWaN) and XXLTraffic evaluation become the next motivated step.

Evidence chain strength: Medium.
Direction is established; causal evidence not yet produced.

Proposed Direction 3: Topology – Persistent Homology

Motivation: correlation dimension estimates show a suggestive pattern:

Dataset	Corr. dim	Panda wins
Weather	0.889	Yes
Lorenz	0.862	Yes
Burgers	0.741	Yes
ETTh1	1.541	No
ETTh2	1.617	No

But the estimator is unreliable on real multivariate data.

Proposed experiment: persistent homology on Vietoris-Rips filtrations of the state-space trajectories for each dataset.

What it would provide:

- Betti numbers ($\beta_0, \beta_1, \beta_2$): connected components, loops, voids
- Persistence diagrams: birth/death of topological features across scales
- Wasserstein distance between Panda training distribution and target dataset topology

If Weather and Lorenz share topological features that ETTh does not: strong motivation for the topological

Connection to previous Suggestions: Current Evidence

Suggestion		What our experiments show	Remarks
G-SWaN / node embeddings		Heterogeneity bottleneck established with 3 controlled experiments. Scalar offset proxy directionally correct but effect too small to be conclusive.	Direction established; no causal evidence yet.
XXLTraffic evaluation		Heterogeneous sensor bottleneck is the correct frame. Traffic sensors are the extreme case of Weather heterogeneity.	Cannot test yet; well-motivated evaluation target.
Topological matching	flow	Topology analysis suggestive: Weather and Lorenz cluster together, ETTh separates. Estimator unreliable. Persistent homology needed.	Pattern exists; evidence quality insufficient to act on.
TimesNet-style training	pre-	Decomposition experiments show both models equally hurt by removing periodicity (Exp 20). Panda not specifically worse on residuals.	No specific evidence this is a bottleneck.

Summary: What This Investigation Established

Three things established with medium-high confidence:

- 1 Panda generalizes beyond chaotic ODEs to Weather and Burgers – but not to ETTh.
- 2 Sensor heterogeneity is a specific architectural bottleneck for Panda on Weather (3 controlled experiments, difficulty and CV matched).
- 3 Channel attention and prediction head are not the positive mechanism and not cheaply recoverable bottlenecks.

One critical gap:

The positive mechanism is unknown. No ablation has causally implicated any component. This requires code access to ablate the Koopman lifting and temporal attention.

Most motivated next step:

Proper node embedding intervention (not scalar proxy) to test whether sensor identity or interaction structure is the heterogeneity bottleneck – and code access for the Koopman lifting ablation.

The Honest Open Question

All experiments show correlations. No experiment has yet produced causal evidence for any single architectural component. Identifying the positive mechanism remains the primary unresolved scientific question.

Thank You

Questions and suggestions welcome.